

5G MU-MIMO EQUALIZER PROTOTYPE

PCB Manual — Handy Edition

Intel 18A proof-of-concept — Vodafone Intelligent Solutions (Open RAN Málaga) · Sasken Silicon

|                |                                                                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document       | PCB Manual — Handy Edition: the 30-page plain-language companion to the PCB Manual v0.9, for newcomers and for presentations                                                                                                                |
| Subject        | What the validation board is for, what the chip needs from it, how each block works and why, what is still unknown, and how the first power-on is run — the same facts and numbers as the full manual, in simple words and with its figures |
| Revision       | 0.9 — follows PCB Manual v0.9 (29 September 2026); open items and assumptions are the full manual's                                                                                                                                         |
| Date           | 29 September 2026                                                                                                                                                                                                                           |
| Prepared by    | Sasken Silicon (compiled with AI assistance from the project document set)                                                                                                                                                                  |
| Basis          | PCB Manual v0.9 and its sources: Programme Progress Review v4 (24 Aug 2026, slides 12–24), Vodafone IP Specifications, System Specification v1.4, Architecture Manual v1.1 (RTL v3.2.2), Physical Design Manual v0.9, vendor datasheets     |
| Readers        | Engineers new to the project or to board design; project leads and Vodafone reviewers who need the picture without the execution detail; anyone preparing a presentation on the board                                                       |
| Classification | Confidential — Vodafone Intelligent Solutions / Sasken Silicon. Not for distribution outside the programme.                                                                                                                                 |

Reading guide

Chapters 1–4 say what the chip is, what the board is for and how a test runs. Chapters 5–12 walk through the board block by block — power, clock, reset, UART, breakout, sockets, layout — each with the figure from the full manual and the reason behind the design. Chapters 13–15 cover simulation, manufacturing and bring-up. Chapter 16 lists the open questions, chapter 17 the words, chapter 18 is a one-page cheat sheet. Wherever the full manual says 'need to ask', this booklet says it too.

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## How to use this booklet

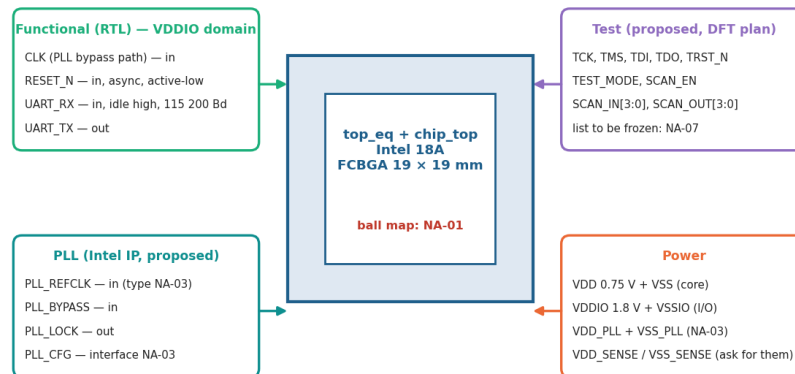
This is the short, plain-language version of the **PCB Manual v0.9** (71 pages). It says the same things with the same numbers, but it leaves out the worksheets, the checklists and most of the source tags. Read it in an hour before a meeting, or use its figures directly in a presentation — every figure here also appears in the full manual, with the same figure content.

| If you want to ...                  | Read here                    | Then in the full manual                                      |
|-------------------------------------|------------------------------|--------------------------------------------------------------|
| understand what the board is for    | chapters 1–4                 | chapters 1–3                                                 |
| explain the power design to someone | chapter 6                    | chapter 6 (worksheets 6.2.1–6.8.1)                           |
| know what is still unknown          | chapter 16                   | chapter 18 (NA-01...NA-20) and chapter 4 (PA-01...PA-14)     |
| draw the schematic or the layout    | chapters 5–12 as an overview | chapters 5–13 — the execution detail is only there           |
| run the first power-on              | chapter 15                   | chapter 16 (stage table, measurement sheet, troubleshooting) |
| look up a word                      | chapter 17                   | Annex A                                                      |

**Key point:** Three markers keep the booklet honest. A plain sentence is a fact from a project document, a datasheet or the RTL. **(assumed)** is a value nobody has confirmed yet. **need to ask** is a question the design cannot be finished without — the full manual lists who answers each one (chapter 18) and what the board does meanwhile.

# 1. The chip, in one page

The Intel 18A device as the board sees it: four functional pins, three supplies, and the pads Sasken adds around the RTL



Sources: Architecture Manual v1.1 §7.2 (pins), §18 (clock/reset), PD Manual v0.9 §2.5 (added pads, rails). Only the four functional pins exist in the RTL; PLL, test and sense pins are proposals until the pad ring and the ball map are frozen. The board must be designed so that a change in the proposed pins costs a footprint edit, not a re-layout: keep the socket escape routing generic (chapter 11).

**Figure 1.** The Intel 18A device as the board sees it: four functional pins, three supplies, and the pads Sasken adds around the RTL.

The chip is a **5G MU-MIMO channel equalizer**: it receives channel data and antenna samples for a block of the radio spectrum (a 'PRB'), solves a matrix problem for it, and returns the equalized symbols. It runs at 1 GHz from a PLL inside the package, and it talks to the outside world through **one serial port (a UART)**. That is all a board designer needs to know about its function; the Architecture Manual has the rest.

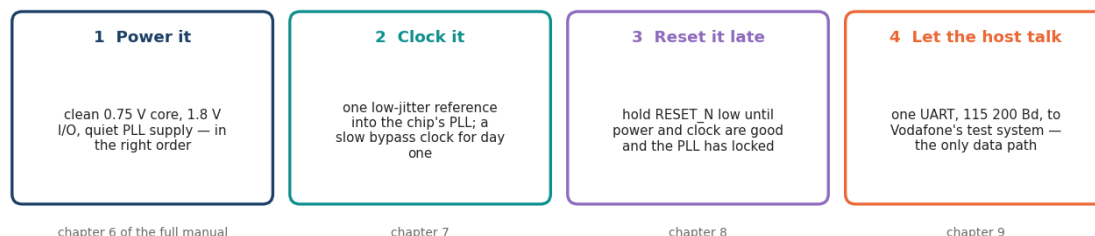
From the board's point of view the chip has **four functional pins** — a clock, an active-low reset, a UART receive and a UART transmit — plus the pins Sasken adds when the chip is built: the PLL reference and its control, a JTAG test port, a few test-mode pins, and the power balls. There is no memory bus, no fast serial link, no analog input. The board's difficulty is not the number of signals; it is a 0.75 V supply that must stay within 3 %, a clean clock into a PLL whose datasheet does not exist yet, and a reset that must be released late and cleanly. (Sources: the RTL itself and System Specification §5.3 for the four pins; the Physical Design Manual for everything marked 'planned', to be frozen in the Pin & Interface Reference, manual 4.)

| Pin                                       | What it is                                                                                                         | What the board does with it                                                                                       |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| CLK                                       | clock input used only in 'bypass' mode (PLL switched off)                                                          | a slow clock for the very first test, through a jumper                                                            |
| RESET_N                                   | reset, active-low, asynchronous                                                                                    | held low until power, clock and PLL are all good; then released with a clean edge                                 |
| UART_RX                                   | serial receive, 115 200 Bd, idle high                                                                              | pulled up to 1.8 V so it never floats; fed from a level translator                                                |
| UART_TX                                   | serial transmit                                                                                                    | to the level translator and the host                                                                              |
| PLL_REFCLK, PLL_BYPASS, PLL_LOCK, PLL_CFG | the PLL's reference clock, bypass switch, 'locked' flag and configuration lines (planned, not yet in any document) | clock buffer → REFCLK; jumper on BYPASS; LOCK into the reset logic; CFG to a header — need to ask (PLL datasheet) |
| TCK, TMS, TDI, TDO, TRST_N + test pins    | JTAG and scan test access (planned)                                                                                | 10-pin debug connector with buffers; test pins on a header                                                        |
| VDD / VSS, VDDIO, VDD_PLL                 | core 0.75 V, I/O 1.8 V (assumed), PLL supply                                                                       | one regulator each, measured and sequenced                                                                        |
| VDD_SENSE / VSS_SENSE                     | two balls this manual asks for                                                                                     | let the regulator measure the voltage <i>at the chip</i> — need to ask (packaging)                                |

## 2. What a validation board is — and its four jobs

When the packaged chips come back from Intel, somebody has to power one up, give it a clock, take it out of reset and send it data — and, when something does not work, see *why*. A **validation board** is the board built for exactly that. It is not a product: it is a laboratory instrument with sockets, test points and LEDs, made in four copies, that lets Vodafone's own test system talk to the chip.

**The board has exactly four jobs — everything in this manual belongs to one of them**



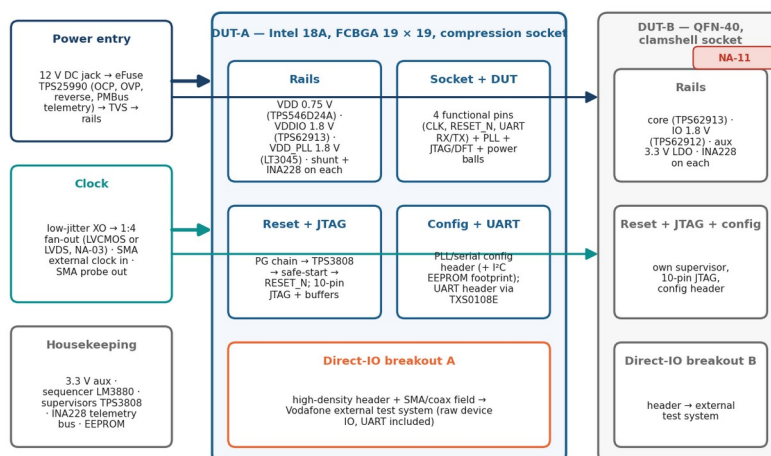
**Plus one job for us, not for the chip: 5 Let us see everything — test points, LEDs, current telemetry on every rail, sockets so that devices can be swapped.**

**Figure 2.** The board has four jobs, plus one for the people using it.

The board hosts **two devices**: DUT-A, the Intel 18A chip in an FCBGA package with a compression socket, and DUT-B, a second device in a QFN-40 package with a clamshell socket, about which the project documents say almost nothing yet (**need to ask**). Each device has its own supplies, reset, JTAG, configuration header and breakout connector; they share the 12 V input and the reference clock. 'DUT' means 'device under test'.

**Why is it built this way?:** The board is deliberately **dumb**. The customer removed the on-board microcontroller, the DRAM and the USB/Ethernet interface from the original concept (24 August 2026). The board only powers, clocks, resets and instruments the chips; Vodafone's external test system does the talking. This makes the bring-up unambiguous: when something fails, it is the device, the socket, a rail, the clock or the reset — and every one of those can be seen on a test point or an LED.

**The validation board at a glance — two independent DUT subsystems, one shared clock, no on-board controller**

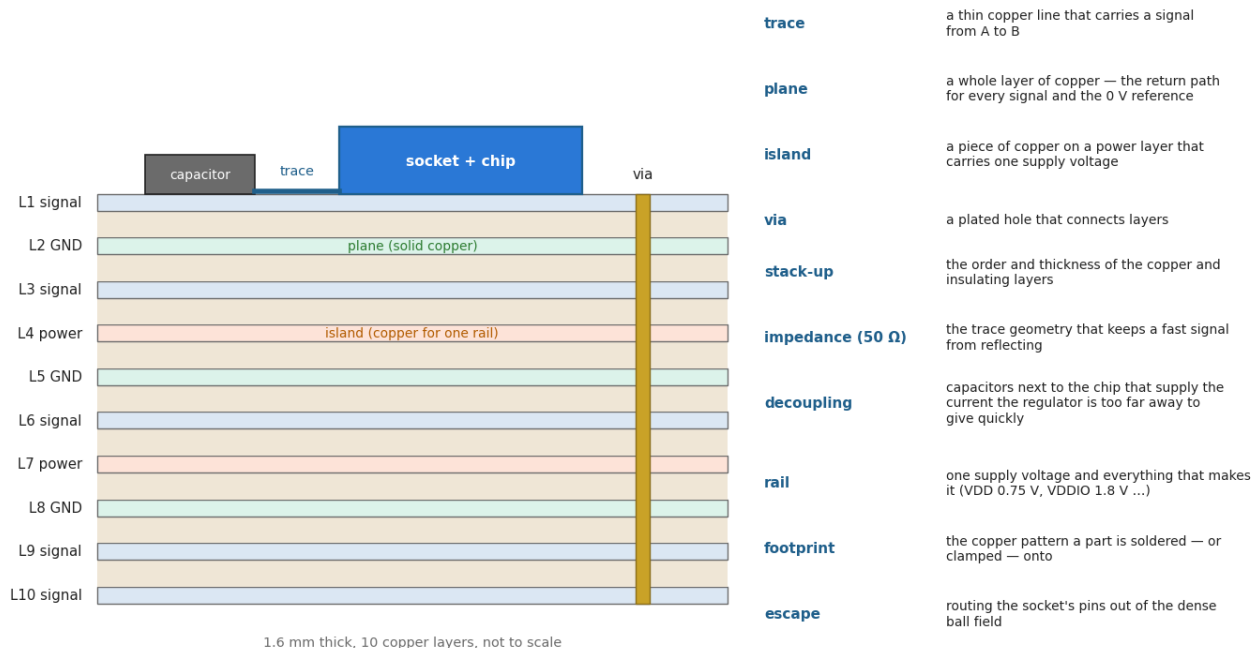


Source: Progress Review v4 (24 Aug 2026), slides 13–15. Everything the external test system needs is on connectors; the board itself only powers, clocks, resets and instruments the two devices. Red tags = items that need an answer before that block can be finished (chapter 18).

**Figure 3.** The whole board at a glance: two device subsystems, one shared clock, no on-board computer.

### 3. Ten words about printed circuit boards

#### What a printed circuit board is made of — the ten words this manual uses all the time



Our board: 10 layers, 1.6 mm, 288 × 216 mm (Progress Review v4, slide 16). Three ground planes, two power layers with islands, five signal layers.

**Figure 4.** What a board is made of, and the words the rest of this booklet uses.

A printed circuit board is a sandwich of copper layers and insulating layers. Copper is shaped into **traces** (signal lines), **planes** (whole layers, used for ground) and **islands** (pieces of a power layer, one per supply). **Vias** are plated holes that connect layers. The order and thickness of the layers is the **stack-up**; ours has ten copper layers in 1.6 mm: five for signals, three solid ground planes and two power layers.

- **Why three ground planes?** Every signal current returns to its source through the nearest ground plane. A signal without a plane next to it radiates, picks up noise and rings. Three planes give every signal layer a neighbour.
- **Why islands instead of power planes?** Each supply needs only a small area; islands keep the 0.75 V copper short and wide, and keep the noisy 12 V away from the sensitive PLL supply.
- **Why 'impedance'?** A fast edge travelling along a trace sees the trace as a transmission line. If the line's characteristic impedance (50 Ω here) does not match the source and the load, part of the edge reflects. On this board only the clock, the SMA lines and the JTAG clock are fast enough to care.
- **Why so many capacitors?** Chapter 6 — the regulator is too far away and too slow to feed a 1 GHz chip's current steps; capacitors next to the socket do it (Figure 15).

## 4. How a test runs

How a test runs: the external host talks to the chip through one UART — everything else on the board is power, clock and reset

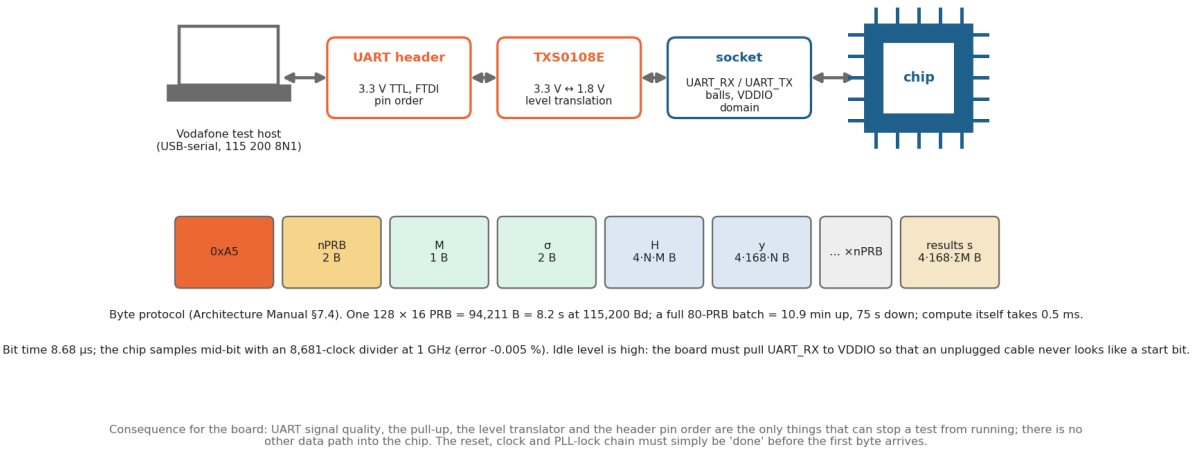
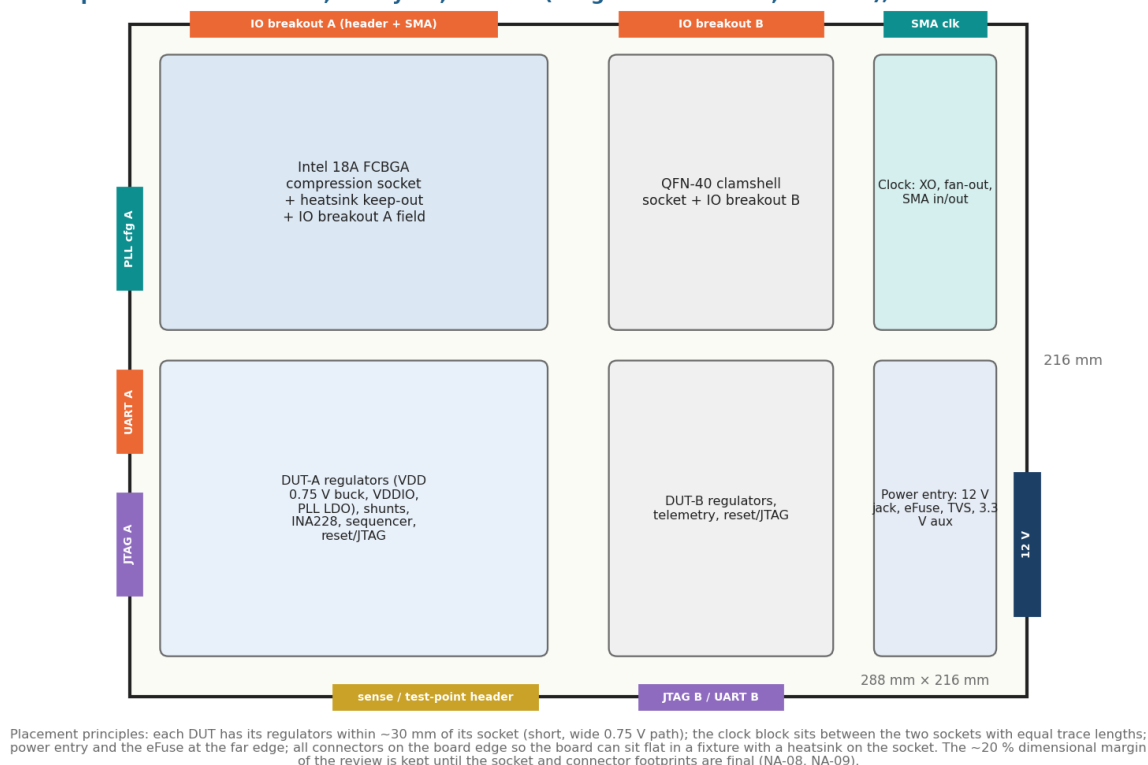


Figure 5. One UART into the chip; everything else on the board is power, clock and reset.

| Phase                                                                                                                                                                                                                                                                    | What happens                                                                                                                         | Board parts involved                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1 Power-up                                                                                                                                                                                                                                                               | 12 V in → protection → 3.3 V housekeeping → 1.8 V I/O → 0.75 V core → PLL supply, one after the other; supervisors confirm each rail | power entry, regulators, sequencer, supervisors |
| 2 Clock                                                                                                                                                                                                                                                                  | the oscillator runs; the buffer drives the PLL reference; the PLL locks and raises PLL_LOCK                                          | clock block, PLL supply                         |
| 3 Reset release                                                                                                                                                                                                                                                          | RESET_N goes high only when all rails are good, the PLL is locked and nobody is holding reset; ≥ 100 ms after the last rail          | reset chain                                     |
| 4 Host connects                                                                                                                                                                                                                                                          | the test PC opens the serial port at 115 200 Bd; the chip's receive line idles at 1.8 V                                              | UART header, translator, pull-up                |
| 5 Upload                                                                                                                                                                                                                                                                 | the host sends 0xA5, the number of blocks, then the data — 8.2 s per block (PRB), 10.9 min for a full batch of 80                    | UART path only                                  |
| 6 Compute                                                                                                                                                                                                                                                                | ≈ 0.5 ms per batch at 1 GHz; the core current jumps with every processing stage                                                      | core rail, decoupling, cooling                  |
| 7 Download                                                                                                                                                                                                                                                               | the results come back — 75 s per batch; the first result byte is the only 'done' signal                                              | UART transmit path                              |
| 8 Repeat                                                                                                                                                                                                                                                                 | the same batch at -3 % and +3 % supply, currents and temperatures logged                                                             | PMBus / telemetry bus                           |
| <b>Watch out:</b> The UART is the <b>only</b> way in and out of the chip. If the UART path has a problem — wrong idle level, swapped pins, a glitch on the receive line — nothing works and nothing tells you why. That is why chapter 9 spends two pages on four wires. |                                                                                                                                      |                                                 |

## 5. The board at a glance

**Board floorplan — 288 × 216 mm, 10 layers, 1.6 mm (Progress Review v4, slide 16); zones and connector edges**



**Figure 6.** Floorplan on the 288 × 216 mm outline: sockets, regulators next to them, the clock between them, connectors on the edges.

Three placement rules explain the floorplan: (1) each device's regulators sit within about 30 mm of its socket, so the 0.75 V copper path is short and wide; (2) the clock block sits between the two sockets, so both reference branches are short and similar; (3) the power entry and the 12 V distribution are at the far edge, away from the PLL supplies and the clock. Every connector is on a board edge. Above each socket there is room for a heatsink; below it, for the socket's backing plate.

**Connector inventory (review slide 15) with the pin-outs proposed in Annex D**

|                  |                           |                                                                                |                                               |
|------------------|---------------------------|--------------------------------------------------------------------------------|-----------------------------------------------|
| <b>J1</b>        | 12 V DC input             | barrel jack 2.5 mm or 4-pin Molex (NA-16)                                      | bench PSU, ≥ 2 A                              |
| <b>J2 / J3</b>   | UART A / UART B           | 6-pin 2.54 mm, FTDI TTL-232R-3V3 order                                         | USB-serial, 115 200 8N1                       |
| <b>J4 / J5</b>   | JTAG A / JTAG B           | 10-pin 2 × 5, 1.27 mm, Cortex-Debug order, keyed                               | debugger / programmer                         |
| <b>J6 / J7</b>   | PLL / device config A / B | 8-pin 2.54 mm: SCLK, SDIO, CS, SDA, SCL, RESET, VDDIO ref, GND (NA-03)         | bench programmer                              |
| <b>J8</b>        | Reference clock in        | SMA, 50 Ω, AC-coupled                                                          | lab clock source                              |
| <b>J9</b>        | Clock / probe out         | SMA, 50 Ω                                                                      | counter, scope                                |
| <b>J10 / J11</b> | 'High-speed' per DUT      | SMA (spare / probe, NA-10)                                                     | bench instruments                             |
| <b>J12</b>       | IO breakout A             | high-density board-to-board (family NA-09), ≥ 40 pins with interleaved grounds | Vodafone test system                          |
| <b>J13</b>       | IO breakout B             | same family (NA-09, NA-11)                                                     | Vodafone test system                          |
| <b>J14</b>       | Sense / telemetry         | 2 × 10 2.54 mm: Kelvin pairs per rail + I <sup>2</sup> C + PG/LOCK/RESET       | DMM, power analyser, I <sup>2</sup> C adapter |
| <b>J15 / J16</b> | Manual reset A / B        | push-button + 2-pin header                                                     | operator / tester                             |

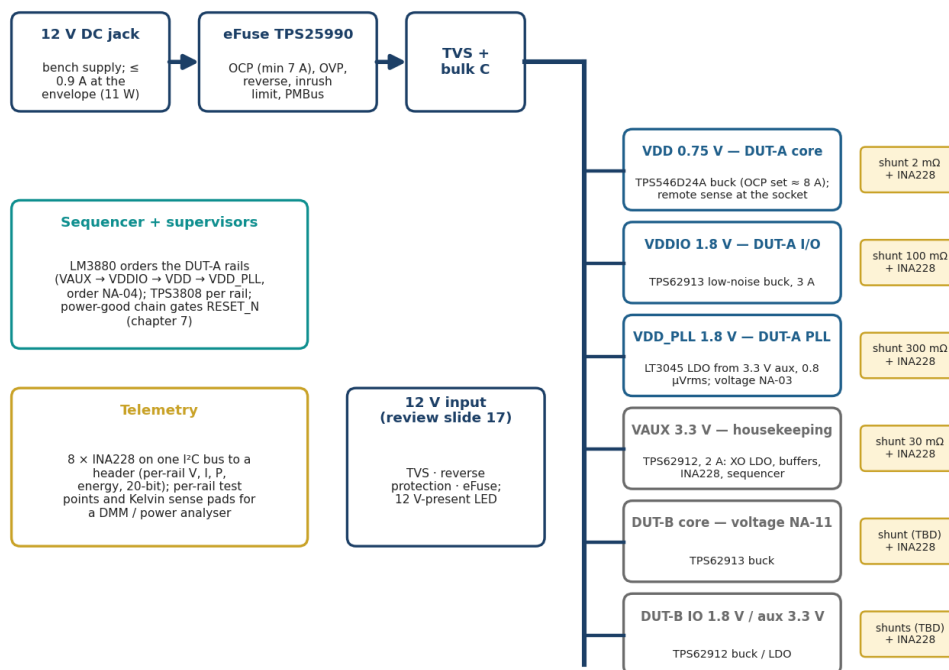
Reference designators are placeholders for the schematic; the connector families for J12/J13 must match Vodafone's test-system cables (NA-09). Every connector is keyed or polarised where a family allows it.

**Figure 7.** Every connector on the board and what plugs into it.

| Block                    | What it contains                                                                                                           | Booklet chapter |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------|
| Power entry              | 12 V jack, protection diode, electronic fuse with telemetry, bulk capacitors                                               | 6               |
| DUT-A rails              | 0.75 V core buck with remote sense, 1.8 V I/O buck, low-noise PLL regulator — each with a current shunt and a monitor chip | 6               |
| Housekeeping             | 3.3 V rail, sequencer, supervisors, the I <sup>2</sup> C bus that reads all the monitors                                   | 6               |
| Clock                    | low-jitter oscillator, 1-to-4 buffer, SMA input, SMA probe output                                                          | 7               |
| Reset / JTAG / config    | safe-start reset logic, 10-pin debug connector with buffers, configuration header, jumpers                                 | 8               |
| UART                     | level translator and the 6-pin header for the USB-serial cable                                                             | 9               |
| Breakout and test points | all device signals on a header at 1.8 V; test points, LEDs, telemetry header                                               | 10              |
| Sockets                  | compression socket for DUT-A, clamshell socket for DUT-B, heatsink provisions                                              | 11              |

## 6. Power — the part that decides whether the board works

**Power tree — one 12 V input, independent instrumented rails per device, every rail through a shunt and an INA228**



Nominal DUT-A draw  $\approx 1.2$  W (core 1 W assumption PA-01); the design envelope is 3 W core / 3.7 W DUT-A so that a pessimistic silicon result still fits. The 40 A core regulator of the review is far above need — acceptable (it brings PMBus telemetry and remote sense), but its current limit must be programmed down so a socket fault cannot sink 40 A.

**Figure 8.** The power tree: one 12 V input, protection, then one regulator per rail, each measured.

### 6.1 The rails

| Rail                | Voltage             | Expected current | Designed for    | Made by                                           | Must stay within                    |
|---------------------|---------------------|------------------|-----------------|---------------------------------------------------|-------------------------------------|
| VDD — core          | 0.75 V              | 1.33 A (assumed) | 4.0 A           | TPS546D24A buck (40 A-class, remote sense, PMBus) | $\pm 3.0\% = \pm 22.5$ mV (assumed) |
| VDDIO — I/O         | 1.8 V (assumed)     | 100 mA           | 300 mA          | TPS62913 low-noise buck                           | $\pm 5\%$                           |
| VDD_PLL             | 1.8 V (need to ask) | 30 mA            | 100 mA          | LT3045 ultra-low-noise LDO                        | $\pm 3\%$ , very quiet              |
| VAUX — housekeeping | 3.3 V               | 350 mA           | 1.0 A           | TPS62912 buck                                     | $\pm 5\%$                           |
| 12 V input          | 12 V                | $\approx 2.4$ W  | 10.6 W → 0.88 A | bench supply through a TPS25990 eFuse             | $\pm 5\%$                           |

Two words explain the table. **Expected** is what we think the chip will draw — about 1 W on the core (assumed; nobody has run the power analysis yet). **Designed for** is three times that, on purpose: the board is built before the chip's power is known, and a board that cannot feed the chip is useless, while a board that can feed three times the chip only costs a bigger inductor. The regulators the review chose are all over-sized for this load; the manual keeps them because they bring things a bring-up board wants — remote sense, telemetry, a programmable output — and it programs their current limits down so that a fault in the socket cannot sink tens of amperes into the chip.

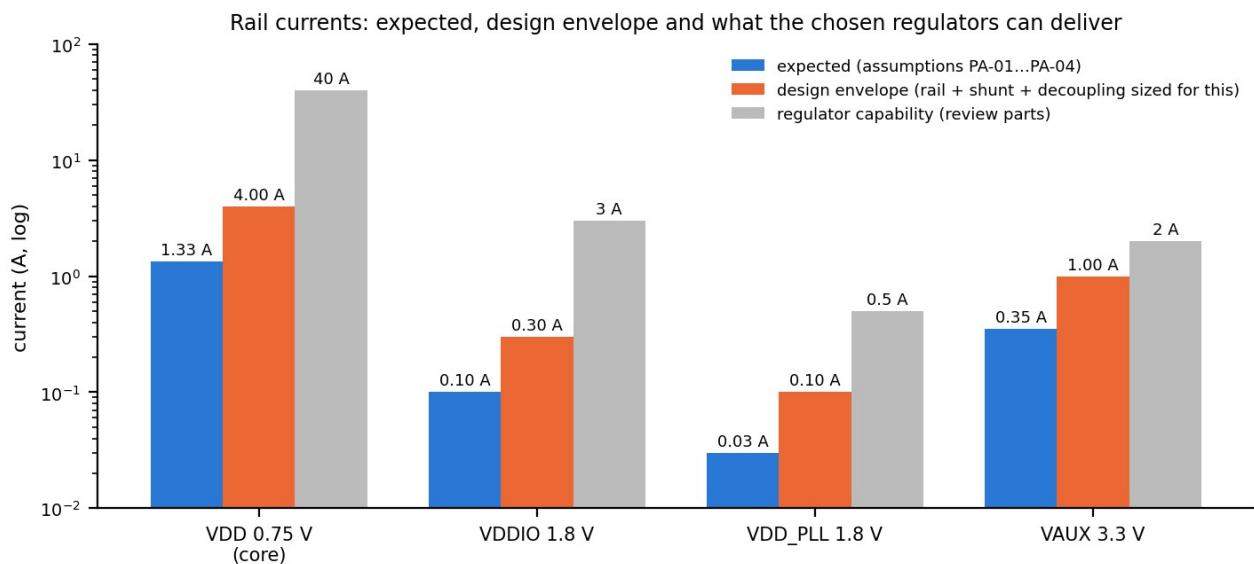
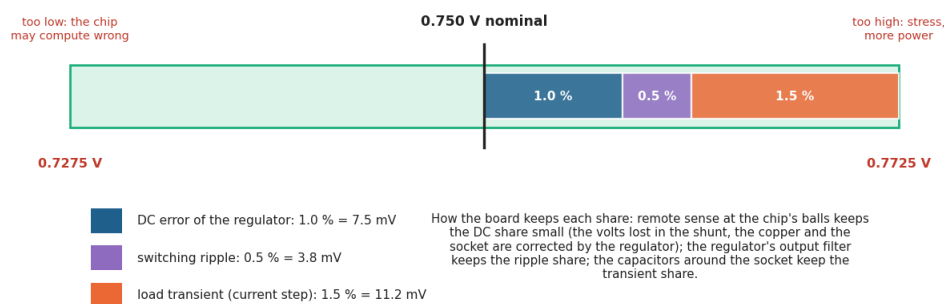


Figure 9. Expected current, design envelope and what each regulator could deliver, per rail.

## 6.2 Why 0.75 V is hard

The core supply must stay within  $\pm 3\%$  of 0.75 V at the chip — that is  $\pm 22.5$  mV, and every effect eats a share

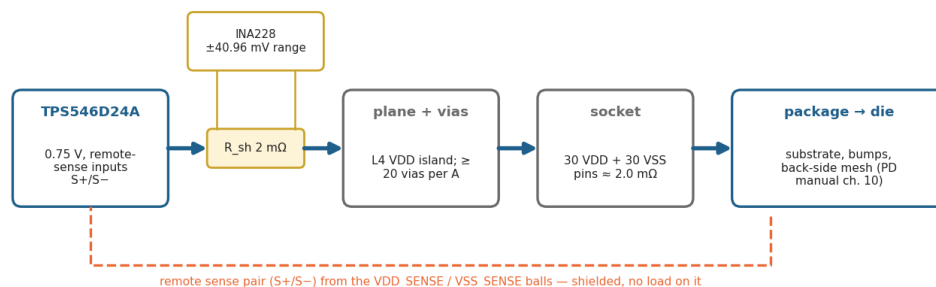


Numbers: numbers\_pcb.py (assumption PA-05: 1 % DC + 0.5 % ripple + 1.5 % transient). The  $\pm 3\%$  itself is an assumption until Intel's operating conditions are known (NA-02).

Figure 10. The  $\pm 3\%$  budget on the core supply and the three effects that share it.

Three percent of 0.75 V is 22.5 millivolts. A 4 A current through the shunt, the copper and the socket loses about that much on its way to the chip — the whole budget, gone before the chip sees anything. The answer is **remote sense**: two extra wires from the regulator to the chip's balls that carry no current and therefore report the true voltage at the chip; the regulator raises its output until the chip, not the regulator, sees 0.75 V.

### The 0.75 V path from the regulator to the die: shunt, remote sense and where the volts are lost



At the 4 A envelope: shunt drop 8 mV (32 mW), socket drop = 8 mV, plane/vias a few mV — all inside the loop when the regulator senses at the balls. Current resolution 39  $\mu$ A per LSB.

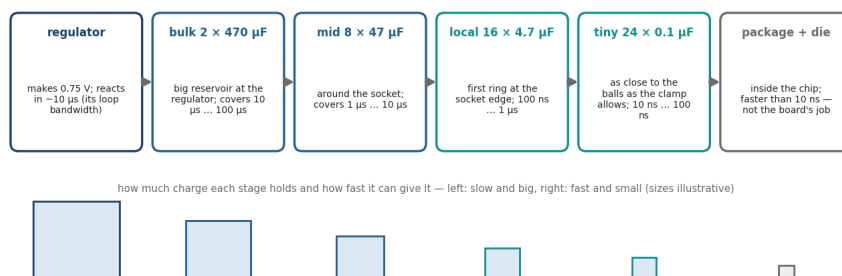
Kelvin rule: the shunt's two sense traces leave from the inside edges of its pads; the INA228 sits within 10 mm; the sense pair is routed together over a ground plane, away from the buck's switch node.

**Figure 11.** Where the millivolts are lost on the way to the chip, and how the remote-sense pair closes the loop at the balls.

**Open item:** The two sense balls do not exist in any document yet. The manual asks the physical-design team and the packaging house for them (**need to ask**). If they cannot be added, two ordinary power balls in the middle of the power cluster are reserved for sensing.

## 6.3 Decoupling — capacitors as reservoirs

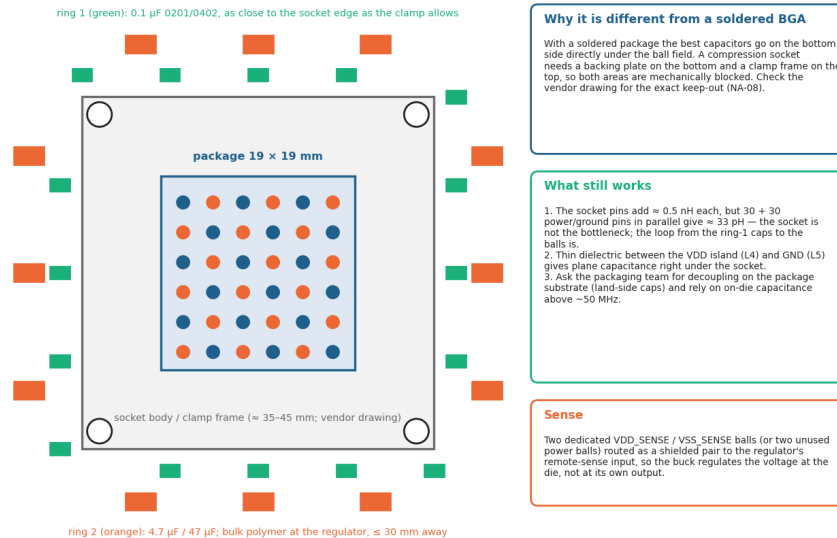
**Why the chip needs capacitors next to it: the regulator reacts in tens of microseconds, the chip changes its current in nanoseconds**



Each stage is a smaller, faster reservoir closer to the chip. The rule that matters: a capacitor only works if its loop to the chip is short — with a compression socket the clamp frame sets that distance, not the layout. The quantities are the starting ladder of PCB Manual §6.6; the post-layout power-integrity simulation finalises them. 'μs' = microsecond (a millionth of a second), 'ns' = nanosecond (a billionth).

**Figure 12.** The regulator is slow, the chip is fast; a ladder of capacitors bridges the gap.

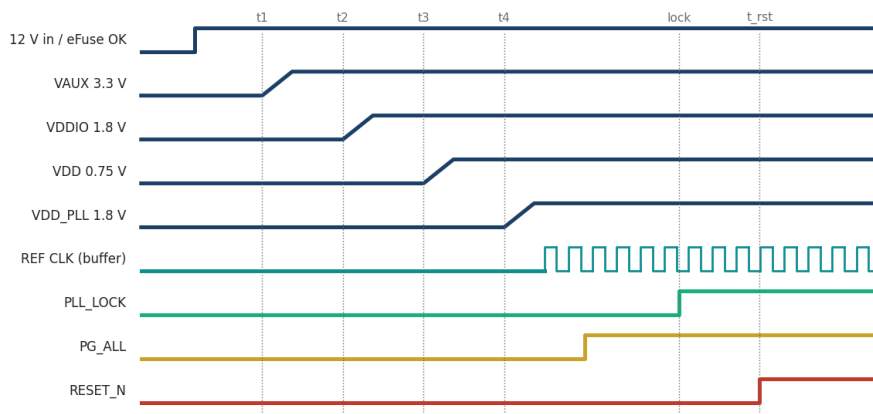
A 1 GHz chip changes its current in nanoseconds. The regulator's control loop reacts in tens of microseconds — a thousand times slower. In between, the current must come from somewhere: from capacitors placed next to the socket, big and slow ones near the regulator, small and fast ones as close to the balls as the socket's clamp allows. The manual derives a **target impedance** — the supply must look like less than 5.6 mΩ to the chip from 100 kHz to about 50 MHz — and a starting ladder of 2 × 470  $\mu$ F, 8 × 47  $\mu$ F, 16 × 4.7  $\mu$ F and 24 × 0.1  $\mu$ F per device. A simulation after layout finalises the quantities.

**Decoupling around a compression socket: the socket body and its backing plate keep capacitors away from the balls — plan for it**

**Figure 13.** A compression socket blocks the best capacitor positions (under the chip); the rings around the socket are the compromise.

## 6.4 Sequencing — the right order, and reset last

### Power-up sequence (proposed) — rails in order, clock before reset, reset released only after PLL lock ('safe start')



**Rules the board implements (sequencer LM3880 + supervisors TPS3808 + one AND gate):**

1. VAUX first (it powers the sequencer, buffers and the PLL LDO). 2. VDDIO before VDD so that no I/O cell is back-driven with its core unpowered — order to be confirmed against the Intel 18A I/O and PLL guides (NA-04).
3. The reference clock runs before RESET\_N is released (the RTL has no clock-present check). 4. RESET\_N = PG\_ALL ∧ PLL\_LOCK ∧ ¬MANUAL\_RESET ∧ ¬TESTER\_RESET, with the supervisor's delay (≈ 100–200 ms) after PG\_ALL.
5. Power-down in reverse order (or all rails together within 10 ms) — the acceptable relaxation must also come from NA-04. 6. Every stage is observable on a test point and an LED; the tester can hold RESET\_N low through the breakout header.

**Figure 14.** Rails come up one after the other; the reset is released only after the last rail is good and the PLL has locked.

The proposed order (assumed — Intel's I/O and PLL guides must confirm it, **need to ask**) is: 3.3 V housekeeping → 1.8 V I/O → 0.75 V core → PLL supply → clock running → reset released. A small sequencer chip (LM3880) enables the regulators 120 ms apart; supervisor chips (TPS3808) watch each rail and pull a common 'power good' line low the moment any rail leaves its window; one AND gate combines power-good, PLL-lock and the manual/tester reset inputs into RESET\_N. The order can be changed with jumpers, so no board re-spin is needed if the answer differs.

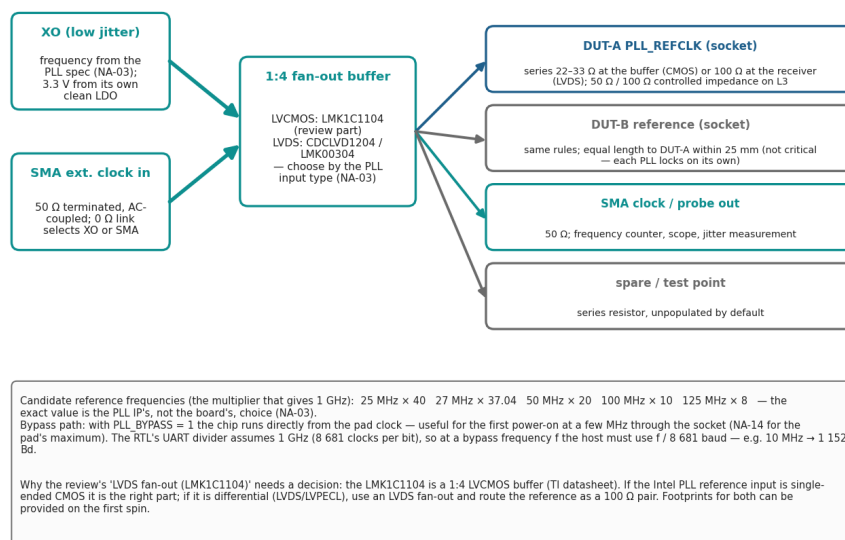
## 6.5 Measuring everything

Every rail passes through a small **shunt** resistor (2 mΩ on the core rail, 30–300 mΩ elsewhere) and an **INA228** monitor that reads the voltage across the shunt with 20-bit resolution — on the core rail that resolves 39.1 μA. All eight monitors sit on one I<sup>2</sup>C bus that comes out on a header, together with the eFuse's and the core regulator's PMBus telemetry. Next to every shunt there is a pair of test points for a multimeter, so the electronics can be cross-checked with a meter at any time.

**New word: shunt** — a very small resistor in series with a supply; the voltage across it is proportional to the current. **Kelvin points** — two test pads placed exactly where you want to know the voltage, so that the meter's wires carry no load current. **PMBus** — an I<sup>2</sup>C-based bus through which the regulator and the eFuse report voltage, current and temperature.

## 7. Clock

**Reference-clock tree — one low-jitter source, a 1:4 fan-out, both devices' PLLs multiply it; an SMA input can replace the oscillator**

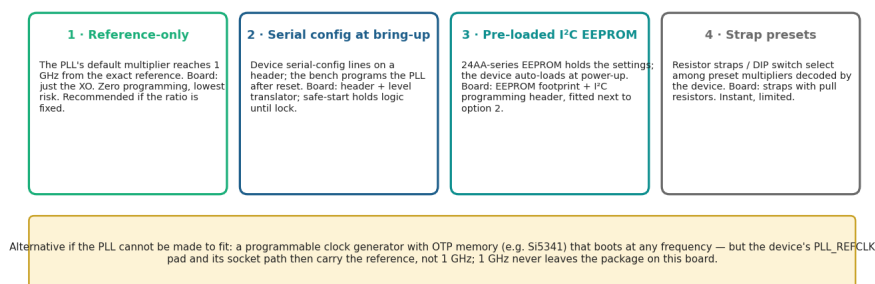


**Figure 15.** One oscillator, one 1-to-4 buffer, two SMA connectors; the chip's PLL makes the 1 GHz inside the package.

The chip's 1 GHz is made inside the package by the PLL; the board only delivers a clean **reference** — a low-jitter crystal oscillator through a fan-out buffer — and, for the first day, a slow **bypass** clock straight into the CLK pad. An SMA input lets a lab signal generator replace the oscillator; an SMA output lets a counter or a scope look at the reference. Frequency, level and signal type of the reference come from the PLL datasheet, which does not exist yet (**need to ask**); the candidates that give 1 GHz with a simple multiplier are 25 MHz, 27 MHz, 50 MHz, 100 MHz, 125 MHz.

**Watch out:** The review calls the buffer an 'LVDS fan-out (LMK1C1104)'. The LMK1C1104 is a **4-output LVCMOS buffer** (TI datasheet) — single-ended, not differential. One of the two statements is a slip, and which one depends on what the PLL's reference input wants. The board carries a footprint for each kind of buffer and populates one when the PLL datasheet arrives. The buffer's output level must also match the pad (1.8 V assumed): a 3.3 V clock into a 1.8 V pad would damage the chip.

**Getting 1 GHz without an on-board controller — the four PLL-programming options of the review, and what each needs on the board**

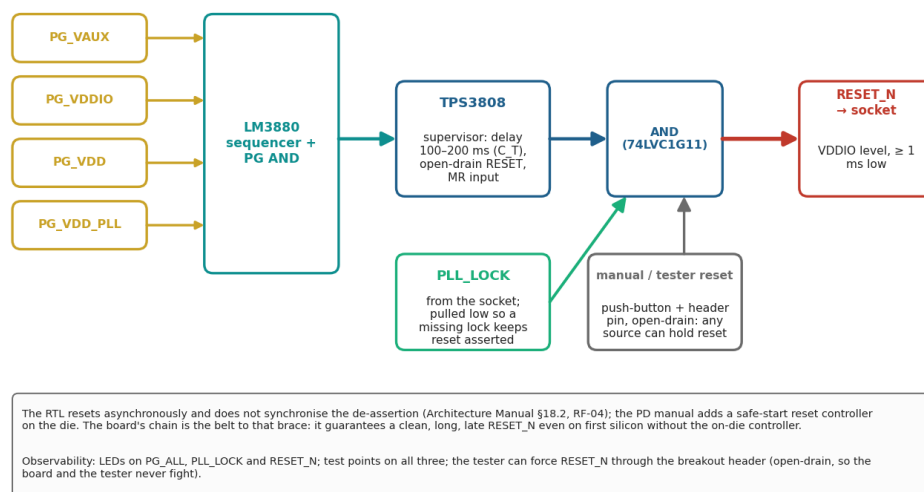


Decision input: the Intel PLL datasheet (NA-03 — reference range, multiplier options, configuration interface). Until it exists the board carries footprints for options 1–3 and the straps of option 4 as DNP.

**Figure 16.** Four ways to get the PLL to 1 GHz without an on-board computer; the board provides for all of them.

## 8. Reset, JTAG and configuration

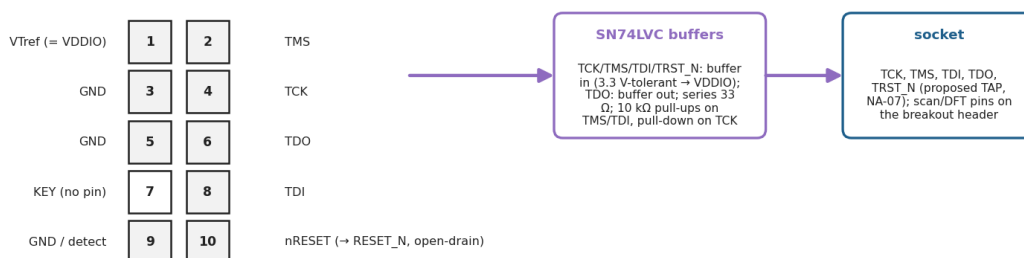
**Reset and safe-start: RESET\_N is released only when every rail is good, the delay has elapsed and the PLL reports lock**



**Figure 17.** The 'safe-start' chain: power good → delay → AND with PLL lock and the manual inputs → RESET\_N.

The chip's reset is asynchronous and the RTL does not synchronise its release; the top level resets only part of its state, and there is no check that the clock is running. None of that is a problem if the board does three things: keep RESET\_N low until every rail is good and the PLL is locked, wait at least 100 ms more, and then release it with one clean edge. A push-button, a header pin and the debugger can all hold reset, but none of them can release it early.

**JTAG / debug: standard 10-pin (2 × 5, 1.27 mm) 'Cortex Debug' pin order per device, with buffers in the VDDIO domain**

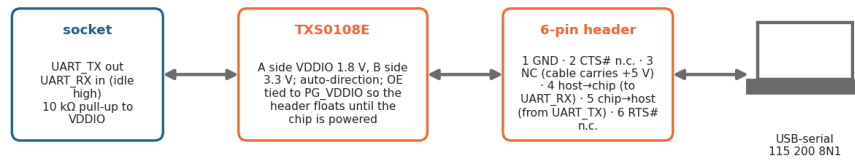


**Figure 18.** The 10-pin JTAG connector in the standard Cortex-Debug pin order, with buffers powered from the chip's 1.8 V.

The JTAG port and the scan-test pins are planned by the Physical Design Manual ('slight DFT'); the board provides the standard 10-pin connector every lab debugger fits, buffers on the five TAP lines, and header positions for the test pins. Which pins exist, and which debugger Vodafone uses, are open questions (**need to ask**). A configuration header with an EEPROM footprint covers every way the PLL might be programmed (serial, I<sup>2</sup>C, straps) until the PLL datasheet says which.

## 9. UART — the only data path

UART path: chip (1.8 V) ↔ level translator ↔ 6-pin 3.3 V TTL header in the FTDI cable order ↔ external USB-serial adapter



Pin order = the FTDI TTL-232R-3V3 cable: pin 4 is the cable's TXD output (host→chip) and goes to the chip's UART\_RX (i\_serial\_rx) through the translator; pin 5 is the cable's RXD input (chip→host) from UART\_TX (o\_serial\_tx). Pin 3 (red wire) carries +5 V from the USB port — left unconnected on the board. Silkscreen by direction, not 'TXD/RXD'.

Protection: 33 Ω series resistors and a TVS array (e.g. 5 V working) on the header side; 1.8 V side ESD is the chip's own I/O cell. Never let 3.3 V reach the socket: the translator's A side must be powered from VDDIO only.

Signal quality: the receiver has one flip-flop and no synchroniser (RF-05) and any byte is a command; keep the RX trace short, referenced to ground, away from the buck switch nodes; add a 100 pF option to ground at the socket for glitch filtering (fitted only if needed).

Translator choice: the review's TXS0108E works for a push-pull UART at 115 kBd; a direction-controlled part (SN74AVC2T245, DIR fixed per line) is the safer alternative if the auto-sensing translator shows edge glitches on the bench — provide the footprint option.

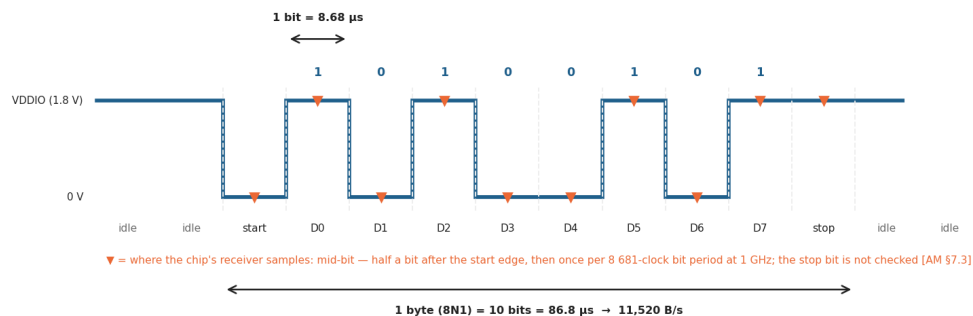
Higher baud rates are an RTL generic (g\_UART\_BAUDS), not a board feature: the header and translator already support ≥ 1 Mb/s.

**Figure 19.** Chip (1.8 V) ↔ level translator ↔ 6-pin header in the FTDI cable order ↔ USB-serial adapter.

The chip's serial lines live at 1.8 V; USB-serial adapters live at 3.3 V; a level translator (TXS0108E) sits between them, and it is switched on only once the chip's 1.8 V is present, so that the cable can never drive an unpowered chip. The header follows the pin order of the common FTDI TTL-232R-3V3 cable, so no adapter is needed.

**Watch out:** Two things about that cable are easy to get wrong. (1) Its **red wire carries +5 V from the USB port**, not 3.3 V — the board leaves that pin unconnected. (2) Its pin 4 is the *cable's* transmit — data going to the chip — so it goes to the chip's receive pin; pin 5 comes from the chip's transmit. The silkscreen therefore says 'host → chip' and 'chip → host', not 'TXD/RXD'.

One UART byte at 115,200 Bd as the scope should show it at the socket test point — example byte 0xA5 (the test-start command)



What to check with the scope (stage 0 loopback, stage 5 with the device): idle level = VDDIO at the socket (not floating, not 3.3 V); a clean falling start edge with no ringing below 0 V or above VDDIO; bit time 8.68 μs ± 2 %; no glitch shorter than a bit on RX while the host is silent (the receiver has no synchroniser — RF-05). In bypass mode the bit time scales with the clock:  $f_{\text{bypass}} / 8\,681$  (10 MHz → 868 μs per bit).

0xA5 = 1010 101₂ is sent LSB first: D0 = 1, D1 = 0, D2 = 1, D3 = 0, D4 = 0, D5 = 1, D6 = 0, D7 = 1. Derived from numbers\_pcb.py (uart block) and the receiver description in Architecture Manual §7.3.

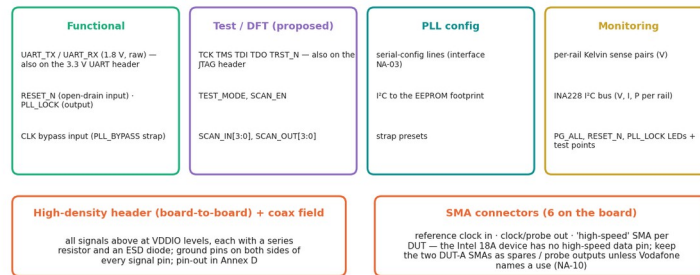
**Figure 20.** One byte at 115 200 Bd as it should look on the scope at the socket.

- One bit lasts 8.68 μs; one byte (start, 8 data bits, stop) 86.8 μs. The chip counts 8,681 clocks of its 1 GHz per bit and samples in the middle of each bit.
- The receive line must **never float and never glitch**: the receiver has no synchroniser and treats every byte as a command. A 10 kΩ pull-up at the socket keeps it high with no cable; 33 Ω series resistors and short routing keep it clean.

- From a slow bypass clock of frequency  $f$  the baud rate becomes  $f / 8,681$  (10 MHz  $\rightarrow$  1 152 Bd) and the host must follow; bring-up uses this on purpose (chapter 15).
- The host streams without waiting — the chip never signals back-pressure. One full-size block takes 8.2 s; a batch of 80 takes 10.9 min up and 75 s down.

## 10. Breakout, test points and protection

### Direct-IO breakout A and instrumentation — everything the external test system and the bench can reach



The review defines the breakout as 'raw functional IO of the device' (slide 15). With four functional pins the header is small; the value of the field is the test access (DFT pins) and the observability (sense, lock, reset), which is why every proposed pad is brought out even if it ends up unused.

**Figure 21.** What the breakout field carries to Vodafone's test system, and what the bench can reach.

Every device signal also appears, raw and at 1.8 V, on a breakout header for Vodafone's external test system — through a 33 Ω resistor and a protection diode, with a ground pin next to every second signal and a reference pin that tells the tester what 1.8 V is. The connector family must match Vodafone's cables (**need to ask**); a plain 2 × 25 header is the placeholder. Six SMA connectors serve the clock in and out and two 'high-speed' lines per device that the Intel chip has no use for (**need to ask** what Vodafone intends).

### What to measure where — test points, headers and expected values on a healthy board

| signal / point                       | expected                                              | instrument                     |
|--------------------------------------|-------------------------------------------------------|--------------------------------|
| VDD core (Kelvin pair at socket)     | 0.75 V ± 3 %, ripple < 5 mVpp                         | DMM / scope 20 MHz BW          |
| VDDIO, VAUX                          | 1.8 V / 3.3 V ± 5 %                                   | DMM                            |
| VDD_PLL                              | 1.8 V (NA-03) ± 3 %, noise < 50 μVrms                 | scope, AC, 20 MHz BW           |
| 12 V after eFuse                     | 12 V; total current ≤ 0.9 A                           | PSU readout / INA228           |
| REF CLK at socket pad (TP)           | frequency per NA-03; edge < 2 ns; jitter per PLL spec | counter / scope 1 GHz BW       |
| SMA probe out                        | = reference (buffer copy)                             | counter                        |
| PG_ALL, PLL_LOCK, RESET_N (TP + LED) | high after the sequence; RESET_N last                 | scope, single-shot at power-up |
| UART_TX / RX at socket (TP)          | idle high (1.8 V), 115 200 Bd frames                  | scope / logic analyser         |
| INA228 bus (header)                  | V, I per rail; core I ≈ 1.3 A expected                | I <sup>2</sup> C adapter       |
| Shunt drop VDD (Kelvin)              | 2 mΩ × I                                              | DMM μV range                   |

Test points: 1 mm pads on a 2.54 mm grid where possible, each next to a ground pad for a probe spring; Kelvin pairs for every rail; all on the top side (the bottom is blocked under the sockets).

**Figure 22.** The measurement map: what to probe, where, the expected value and the instrument.

### Protection at the board edge — every external pin gets the same three things: a series resistor, a clamp and a defined idle level

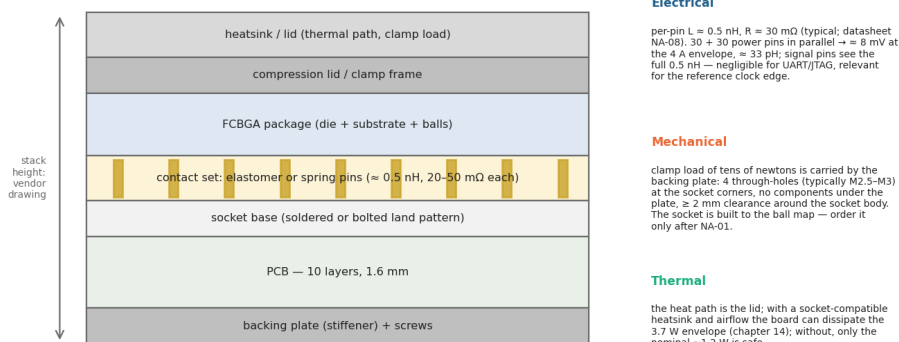
|                                     |                                                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>12 V jack</b>                    | TVS (SMB) class, 15 V, reverse-polarity FET or eFuse feature, bulk + ceramic; inrush limited by the eFuse                                                                         |
| <b>UART / JTAG / config headers</b> | 33 Ω series; low-capacitance TVS array (≤ 5 pF) at the header; pull-up/down to the idle level on the chip side; translators powered from VDDIO only                               |
| <b>SMA clock in/out</b>             | DC-block capacitor; 50 Ω termination; ESD diode rated for RF (< 1 pF); never drive the socket pad from the SMA without the buffer                                                 |
| <b>Breakout header</b>              | series resistor + ESD diode per signal; ground pins interleaved; VDDIO reference pin so the tester can level-shift; no 3.3 V on any DUT signal pin                                |
| <b>Sense / telemetry headers</b>    | 1 kΩ series on the Kelvin lines to J14 (DMM only, no current — protects the rails from a mis-plugged cable); INA228 I <sup>2</sup> C with pull-ups on the board; keyed connectors |

The chip's own I/O cells carry the on-die ESD (Intel 18A GPIO, PD manual A9); the board protection exists so that the lab handling of a socketed, swappable device never reaches them. Rating targets: IEC 61000-4-2 contact ±8 kV on external connectors (assumption PA-12; confirm with the lab's own requirement, NA-18).

**Figure 23.** Every external pin gets the same three things: a series resistor, a clamp and a defined idle level.

## 11. Socket, package and the second device

### Compression socket cross-section — what sits between the ball and the board, and what it costs electrically and mechanically

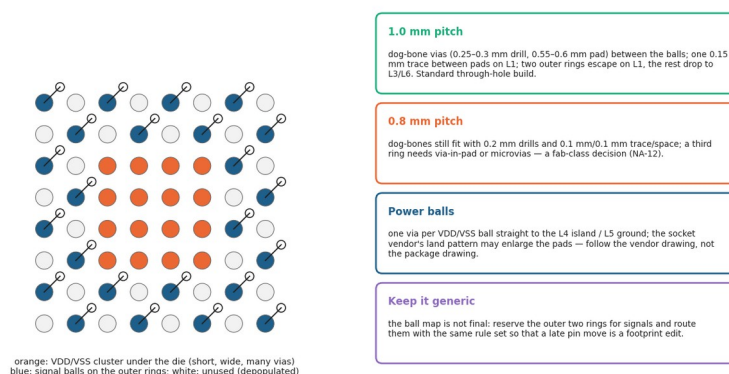


Sockets named in the review: Ironwood GHZ-BGA (elastomer, high bandwidth) or a Plastronics spring-pin socket for the FCBGA; Plastronics 40QN50S clamshell (0.5 mm pitch) for the QFN-40. Final part after the ball map (NA-01, NA-08).

**Figure 24.** A compression socket: what sits between the ball and the board, and what it costs electrically and mechanically.

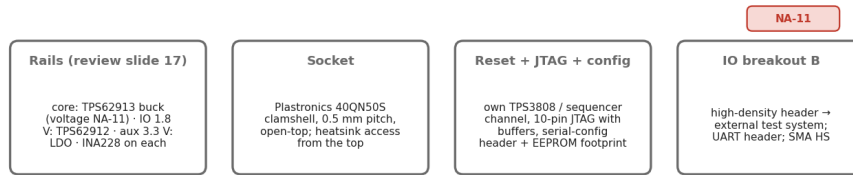
The chip is not soldered: a **compression socket** presses its balls onto spring pins (or an elastomer) with a clamp lid, so devices can be swapped and a damaged one does not cost a board. The price is a longer path for the current (about  $30 \text{ m}\Omega$  and  $0.5 \text{ nH}$  per pin — assumed until the vendor's data arrives), a blocked area under the chip where the best capacitors would go, and a backing plate on the bottom side. The socket is built to the chip's **ball map** — which does not exist yet: the packaging house derives it from the chip's bump map once the pad ring is designed (**need to ask**, the single most important open item for the board).

### Escape routing under the socket land pattern — dog-bone fan-out at an assumed 1.0 mm pitch (NA-01); depopulated map, power balls clustered



**Figure 25.** 'Escape' routing: getting the socket's pins out of the ball field — signals on the outer rings, power in the middle.

To keep going without the ball map, the manual proposes a *pattern* rather than positions: power and ground balls in the centre, the two sense balls inside that cluster, the PLL in one corner, and all  $\sim 25$  signals on the outer two rings. If the final map follows the pattern, the socket footprint is a small edit; if it does not, the escape routing changes. The board also asks for one more thing the chip does not need but the board does: that every outer-ring position gets a via and a route, used or not, so that a late pin move is a jumper change, not a re-layout.

**DUT-B subsystem (QFN-40 device) — mirrors DUT-A with its own rails, reset and JTAG; everything device-specific is open (NA-11)**

What is known: the review's architecture (independent power domains, shared reference clock, direct IO), the socket part and the regulator families. What is not: the device identity (the IP spec mentions a TSMC 12 nm variant of the same IP with a QFN-20 package — the review says QFN-40), its pin-out, core voltage, I/O levels, clock/PLL needs and JTAG. Until NA-11 is answered the DUT-B section of the schematic is a copy of DUT-A with the QFN socket and generic rail set-points, and its bring-up procedure is chapter 15 applied twice.

**Figure 26.** DUT-B as far as the review defines it; everything device-specific is open.

DUT-B is a QFN-40 device in a Plastronics clamshell socket with its own regulators, reset, JTAG, configuration header and breakout. The project documents never name the device (the IP specifications mention a TSMC 12 nm version of the same design, which may be it). Until Vodafone answers, DUT-B's schematic is a copy of DUT-A's with every device-specific value left as a resistor option (**need to ask**).

## 12. Stack-up and layout

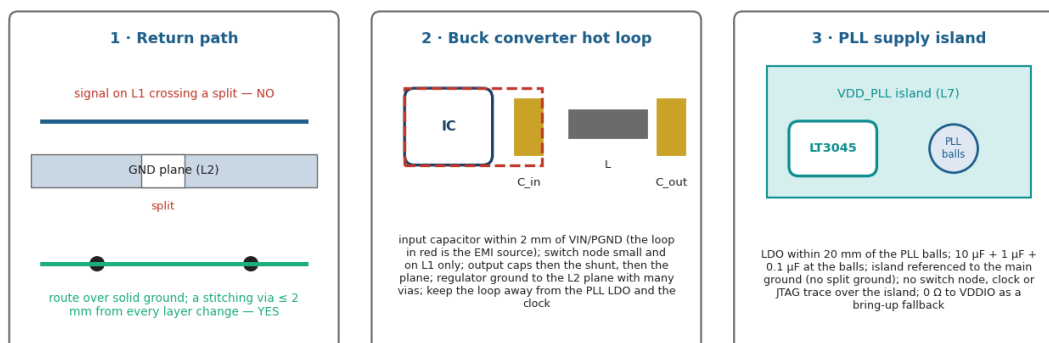
10-layer stack-up (review slide 16) with the proposed assignment; final dielectric thicknesses come from the fabricator's impedance calculation

|     |                                                                                             |      |                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1  | Signal — top: socket escape, components, 50 $\Omega$ microstrip                             | 1 oz | Material: high-Tg FR-4; a low-loss laminate only where a fast edge exists (reference clock) — the review's 'HS laminate on critical nets'.                              |
| L2  | GND                                                                                         | 1 oz |                                                                                                                                                                         |
| L3  | Signal — controlled impedance: reference clock (50 $\Omega$ / 100 $\Omega$ ), JTAG, UART    | 1 oz | Impedance: 50 $\Omega$ single-ended, 100 $\Omega$ differential (review); ask the fab for a stack-up with these targets on L1 and L3 and impedance coupons on the panel. |
| L4  | PWR — VDD 0.75 V islands (DUT-A, DUT-B core), thin dielectric to L5                         | 2 oz |                                                                                                                                                                         |
| L5  | GND                                                                                         | 1 oz | Every signal layer has an adjacent ground (L2, L3, L8); no signal crosses a plane split without a return via nearby.                                                    |
| L6  | Signal — general, telemetry, control                                                        | 1 oz |                                                                                                                                                                         |
| L7  | PWR — VDDIO 1.8 V, VDD_PLL, VALUX 3.3 V, 12 V islands                                       | 2 oz | Power planes are islands, not full planes: each DUT gets its own 0.75 V island on L4, sized to the current with 3 oz copper; 12 V is a narrow island near the entry.    |
| L8  | GND                                                                                         | 1 oz |                                                                                                                                                                         |
| L9  | Signal — general                                                                            | 1 oz | Thickness 1.6 mm, ENIG finish (socket contact + fine pitch). Via class: through-hole assumed; microvias only if the ball pitch forces them (NA-12).                     |
| L10 | Signal — bottom: test points, sense routing, DNP options (no parts under the backing plate) | 1 oz |                                                                                                                                                                         |

**Figure 27.** The review's 10-layer stack-up with the layer assignment the manual proposes.

The review fixes the stack-up: 10 layers, 1.6 mm, high-temperature FR-4 with a low-loss laminate where the clock runs, 50  $\Omega$  / 100  $\Omega$  controlled impedance, gold (ENIG) finish for the socket contacts. The manual adds what goes on each layer: socket escape and fast lines on L1 and L3, the 0.75 V islands on L4 next to the L5 ground plane (thin dielectric = free capacitance), the other rails on L7, sense pairs and control on L6, ground on L2, L5 and L8.

**Layout rules that matter on this board — return paths, plane islands, the buck loops and the PLL island**



General: 0.75 V distribution as a plane island with  $\geq 20$  vias per ampere at every layer change; shunt and sense as in chapter 5; all connectors on the edge with their ESD parts within 5 mm; test points  $\geq 1$  mm on a 2.54 mm grid where possible; no components inside the socket keep-out on either side.

Signals: reference clock 50  $\Omega$  / 100  $\Omega$  on L3 with ground on L2 and L4 side (L4 island under the clock = avoid → route the clock over L2/L5 stripline where the L4 island is absent); UART/JTAG  $\leq 100$  mm, series-terminated, no stubs; length-matching is not required on this board except the clock branches ( $\leq 25$  mm).

**Figure 28.** The three rules with the biggest effect on this board: return paths, the buck's hot loop, and the PLL's supply island.

## 12.1 Ten layout rules, in order of importance

1. Sockets first, from the vendor's drawings, with their keep-outs, holes and heatsink clearance; nothing else moves them later.
2. Regulators next to their loads: the 0.75 V buck within 30 mm of the socket edge, inductor and capacitors between the chip and the socket, the shunt at the island's entrance.
3. Every signal needs a ground plane next to it, all the way; never cross a split, never route the clock over a power island.
4. The 0.75 V island is 2 oz copper, at least 10 mm wide, uncut; the shunt is the only path (no copper or vias around it).
5. Switching parts (the bucks' switch nodes) at least 20 mm from the oscillator, the buffer, the PLL regulator and the sense pairs.
6. The remote-sense pair: a tightly coupled pair over solid ground, guarded, no layer changes, only the datasheet's own filter parts at the regulator.
7. Capacitors: two vias per capacitor, side by side, as close to the socket edge as the clamp allows; bulk ones at the regulator.
8. Clock: on L3 between two grounds, 50  $\Omega$  (or 100  $\Omega$  pair), under 150 mm, both branches within 25 mm of each other, series resistor at the buffer, no stubs.
9. Connectors on the edges with their protection parts within 5 mm; the UART receive line short and away from the bucks.
10. Silkscreen that a stranger can use: every connector name and pin 1, every jumper's default, every test point's net and voltage, the board serial number.

### Layout work package — the order that avoids re-work on a socketed, power-critical board (chapter 13.9)

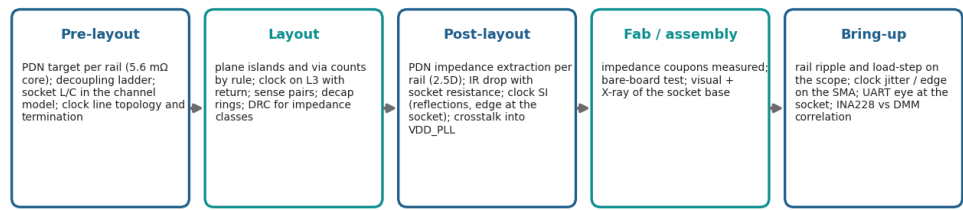


Steps 2 and 8 are the ones that must wait for NA-01/NA-08 (ball map, socket drawing); everything else can be done with the placeholder footprint and re-checked when the drawings arrive. Step 12 feeds the layout review checklist (chapter 17.2).

**Figure 29.** The layout work package: twelve steps in the order that avoids re-work.

# 13. Checking the design before it is built — and keeping it cool

Signal- and power-integrity work on this board — what is simulated, when, and what is measured to close the loop



There is no multi-gigabit link on this board; 'SI' here means the reference clock edge, the UART/TAG cleanliness and the return paths. 'PI' is the real work: a 0.75 V rail with a 3 % budget through a socket. Tool-independent: any 2.5D PDN extractor and any SPICE with the socket vendor's model will do (NA-12).

Figure 30. What is simulated when, and what is measured afterwards to close the loop.

There is no multi-gigabit link on this board, so 'signal integrity' means only: the shape of the reference clock at the socket, the cleanliness of the UART receive line, and the return paths. The real analysis is **power integrity**: does the 0.75 V network stay below its target impedance from the regulator to the socket, does the DC drop match the budget, does a 2 A current step stay inside 11 mV? These are simulated after layout with the socket vendor's model, and then measured at bring-up to see whether the model was right.

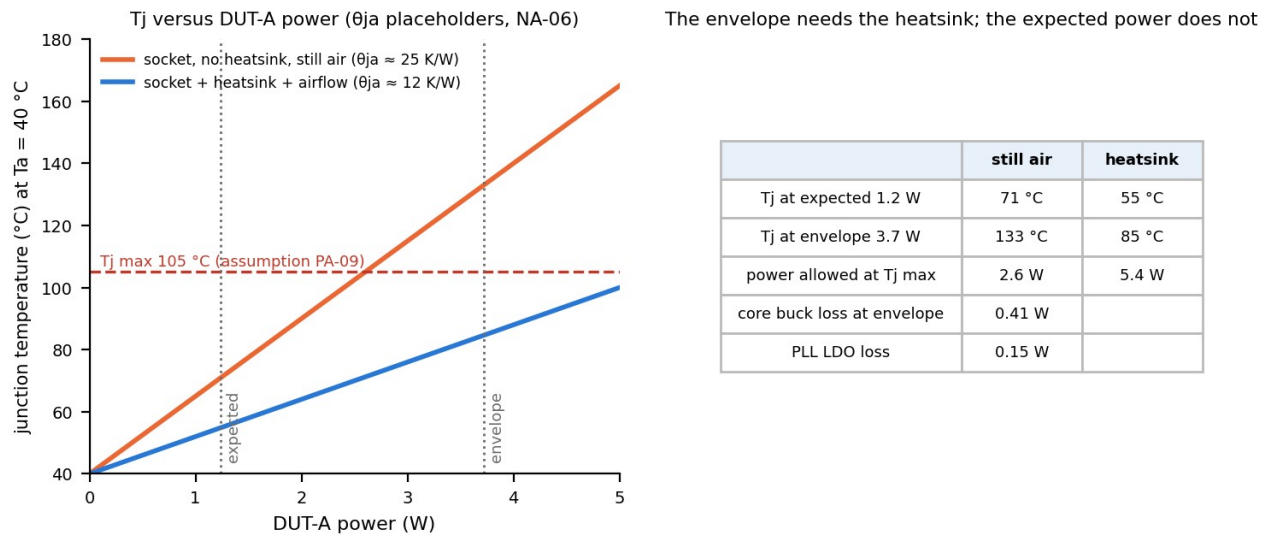
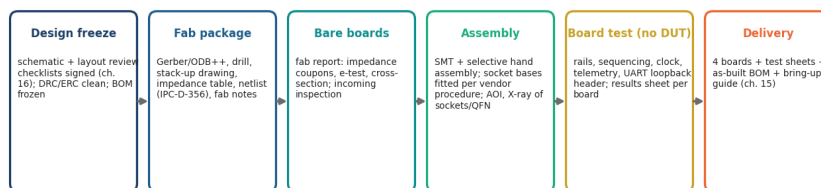


Figure 31. Junction temperature versus chip power with placeholder thermal resistances.

With placeholder numbers (assumed until the package's thermal data arrives, **need to ask**) the socketed chip reaches about 71 °C at its expected 1.2 W in still air — fine — and about 133 °C at the 3.7 W design envelope — not fine. The board therefore provides for a heatsink on the socket lid and a fan bracket; the bring-up sheet records the lid temperature at every power level.

## 14. From design files to four tested boards

From design freeze to four tested boards — the release chain and the checks at each hand-over

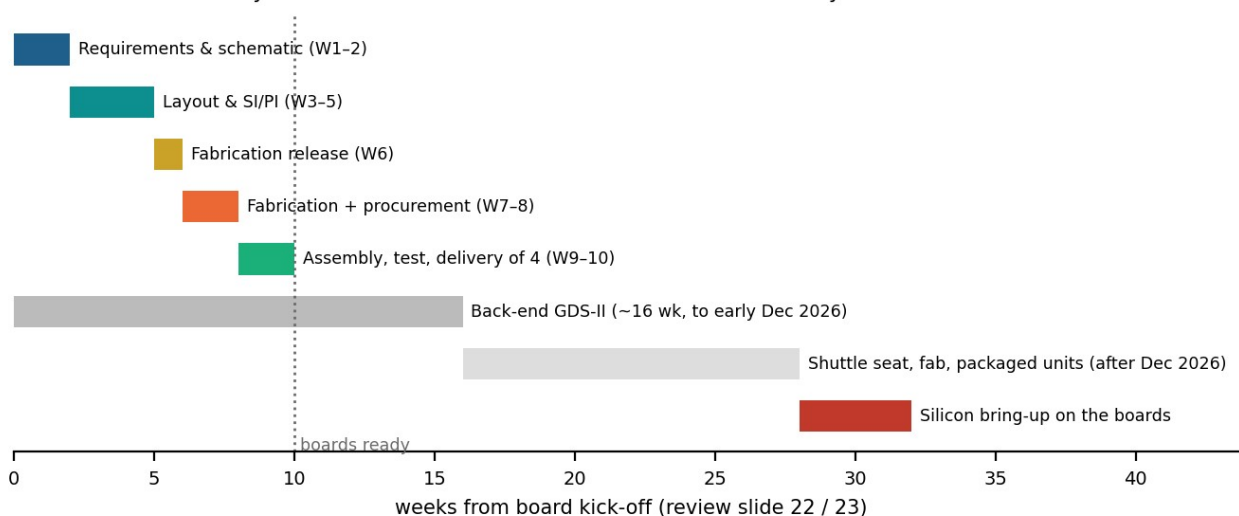


Review plan (slide 22): W1-2 requirements & schematic · W3-5 layout & integrity · W6 fabrication release · W7-8 fabrication and procurement · W9-10 assembly, test and delivery of 4 units. One design iteration is included in the scope (slide 13).

**Figure 32.** From design freeze to four tested boards.

The fabricator receives the layout files, the drill file, the stack-up drawing with the impedance targets and a numbered list of fabrication notes; it returns bare boards with an impedance report, a cross-section and an electrical test. The assembler solders everything except the sockets and connectors, which are fitted afterwards by hand or per the socket vendor's procedure. Every board then passes the same **board-level test without a chip**: rails at their set-points, the power-up order on a scope, telemetry readback, the clock at the socket, a UART loopback. Only then does a chip go in.

The board is ready ~20 weeks before silicon: use the time for board-only tests and the 'need to ask' items

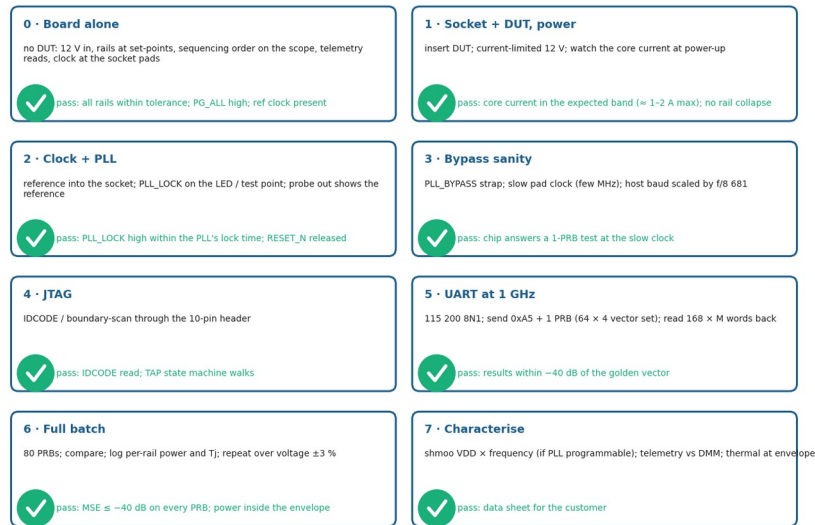


**Figure 33.** The review's 10-week plan for the board against the silicon schedule.

The review includes one design iteration. The manual's footprint options (two clock buffers, two translators, EEPROM, jumper-selectable sequencing) and the generic escape routing are meant to spend that iteration on the things that cannot be foreseen — the final ball map and the PLL's needs — rather than on the things that can.

## 15. Bring-up in eight stages

### Silicon bring-up — eight stages, each with a pass criterion; never move on with a failed stage



The stages use only what the board provides: bench PSU, scope, frequency counter, USB-serial adapter, JTAG probe, the INA228 bus. Stage 3 exists so that a dead PLL does not block functional testing. Details and the troubleshooting table are in chapter 15.

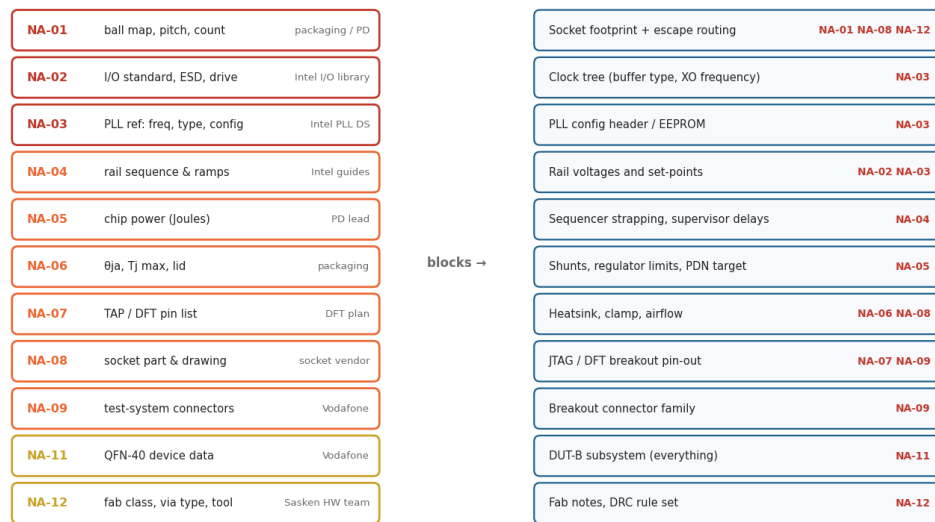
**Figure 34.** Eight stages, each with a pass criterion; a stage is never skipped and never repeated with a failed predecessor.

| Stage            | What you do                                                                                                                                                           | Pass when                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 0 Board alone    | power the board without a chip; measure every rail; record the power-up order; read all monitors; probe the clock at the socket; UART loopback through the translator | rails inside their windows; order as designed; clock present; loopback bytes echo            |
| 1 Chip in, power | insert the chip per the socket procedure; power on with a 1 A supply limit; watch the core current                                                                    | current in the expected band; no rail collapses; no fuse trip; reset still low (no lock yet) |
| 2 Clock and PLL  | reference from the oscillator; watch the PLL-lock LED and the reset release                                                                                           | PLL_LOCK high; RESET_N released after the delay                                              |
| 3 Bypass sanity  | PLL bypassed, 10 MHz into the CLK pad, host at 1 152 Bd; send one small block                                                                                         | correct results (checked against the MATLAB reference)                                       |
| 4 JTAG           | debugger on the 10-pin connector at 1.8 V; read the ID code                                                                                                           | the test port answers                                                                        |
| 5 UART at 1 GHz  | PLL mode; 115 200 Bd; one full-size block                                                                                                                             | correct results in 8.2 s                                                                     |
| 6 Full batch     | 80 blocks; compare every result; log current and temperature; repeat at $-3\%$ and $+3\%$ supply                                                                      | all correct at all three voltages; current and temperature inside the envelope               |
| 7 Characterise   | supply × frequency sweep, power per phase, telemetry vs meter, thermal with the heatsink, an eight-hour run                                                           | a data sheet of the chip on this board                                                       |

**Mentor's note:** Before the first power-on with a chip: socket contacts inspected with a magnifier, all jumpers at their defaults, LOCK\_FORCE jumper removed, supply current limit set, heatsink not yet fitted, and two people present. Everything measured goes into the board's test sheet with the board and chip serial numbers.

## 16. What we still need to ask

What blocks what — the open questions and the design steps that cannot be finished without them



Red: cannot order the socket or finish the footprint without it. Orange: affects schematic values or straps (footprints can be placed now). Gold: separate subsystem / process choice. Full register with owners in chapter 18.

**Figure 35.** Which open question blocks which design step.

Twenty questions are open at rev 0.9; the full manual (chapter 18) lists all of them with who answers, what they block and what the design does meanwhile. The ten that matter most:

| #     | Question                                                                                            | Who answers                     | Until then                                                     |
|-------|-----------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------|
| NA-01 | The chip's ball map — and can two sense balls be added?                                             | packaging house via the PD lead | generic escape; socket order on hold                           |
| NA-02 | Intel 18A I/O: voltage (1.8 V?), thresholds, 3.3 V tolerance, ESD rating                            | PD lead (I/O library)           | 1.8 V CMOS, not 3.3 V tolerant                                 |
| NA-03 | The PLL datasheet: reference frequency, type (CMOS/LVDS), level, supply, configuration, lock output | Intel via the PD lead           | both buffer footprints; header covers SPI and I <sup>2</sup> C |
| NA-04 | Required power-up / power-down order and delays                                                     | Intel integration guides        | assumed order; jumpers allow any order                         |
| NA-05 | The chip's power at 1 GHz                                                                           | PD lead (power analysis)        | 1 W expected, 3 W envelope                                     |
| NA-06 | Package thermal data and clamp-pressure limits                                                      | packaging house                 | placeholders; heatsink provision                               |
| NA-07 | The DFT pin list and Vodafone's debugger                                                            | DFT plan + Vodafone             | Cortex 10-pin order; header positions reserved                 |
| NA-08 | Socket part number, drawing, electrical model, torque                                               | socket vendor after NA-01       | placeholder footprint and parasitics                           |
| NA-09 | Vodafone's test-system connectors, levels and cables                                                | Vodafone                        | 2 × 25 header; FTDI/Cortex pin orders                          |
| NA-11 | What DUT-B is                                                                                       | Vodafone                        | copy of DUT-A with resistor options                            |

The manual also carries fourteen **assumptions** (PA-01 ... PA-14) — the chip's power, the rail tolerances, the power-up order, the socket parasitics, the thermal resistances, the ball pitch, the reference type, the ESD target, the connector orders and DUT-B's similarity to DUT-A — each with an owner and a way to confirm it, and forty datasheet values marked '[DS-check]' that the schematic engineer reads from the datasheets when the sheets are drawn.

## 17. Words

| Word                                        | Meaning here                                                                                                                        |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| DUT                                         | device under test — DUT-A is the Intel 18A chip, DUT-B the QFN-40 device                                                            |
| FCBGA                                       | flip-chip ball-grid array — a package with solder balls underneath; ours is 19 × 19 mm                                              |
| QFN                                         | quad flat no-lead — a small package with pads along its edges; DUT-B's package                                                      |
| compression socket / clamshell socket       | sockets that press the package onto contacts with a lid — no soldering                                                              |
| rail                                        | one supply voltage with its regulator: VDD (core, 0.75 V), VDDIO (1.8 V), VDD_PLL, VAUX (3.3 V)                                     |
| buck / LDO                                  | two kinds of regulator: a switching one (efficient, a little noisy) and a linear one (quiet, wastes the voltage difference as heat) |
| eFuse                                       | an electronic fuse: a chip that switches the input on slowly and off quickly on a fault, and reports what it sees                   |
| envelope                                    | the current or power a rail is designed to deliver — larger than the expected value on purpose                                      |
| remote sense / Kelvin                       | measuring a voltage at the load with two wires that carry no current                                                                |
| shunt, INA228                               | a small series resistor and the chip that reads the voltage across it to know the current                                           |
| decoupling, PDN, target impedance           | the capacitors that feed the chip's fast current steps; the whole power path; the impedance it must stay below                      |
| sequencer, supervisor, power good (PG)      | the chip that switches rails on in order; the chip that watches a rail; the signal that says a rail is inside its window            |
| safe start                                  | releasing reset only after all rails are good, a delay has passed and the PLL reports lock                                          |
| PLL, reference, bypass                      | the circuit that multiplies a slow clock to 1 GHz; the slow clock it multiplies; running the chip from the slow clock directly      |
| jitter                                      | the timing noise of a clock edge — small is good; the PLL's datasheet says how small                                                |
| UART, baud, 8N1                             | the serial port; bits per second (115 200 here); 8 data bits, no parity, 1 stop bit                                                 |
| level translator                            | a chip that converts 1.8 V signals to 3.3 V and back                                                                                |
| JTAG, TAP, DFT, scan                        | the test port and the on-chip test structures the PD team plans                                                                     |
| stack-up, plane, island, via, trace, escape | chapter 3                                                                                                                           |
| impedance-controlled (50 Ω / 100 Ω)         | traces whose geometry is set so fast signals do not reflect                                                                         |
| SI / PI                                     | signal integrity / power integrity — the two kinds of simulation in chapter 13                                                      |
| DRC, ERC                                    | the layout tool's and the schematic tool's automatic rule checks                                                                    |
| Gerber / ODB++, BOM, DNP                    | the files the fabricator needs; the parts list; 'do not populate' — a footprint left empty on purpose                               |
| bring-up                                    | the first power-on and the stage-by-stage tests that follow                                                                         |
| NA-xx, PA-xx, BR-xx, [DS-check]             | need-to-ask item, assumption, board requirement, datasheet value still to be read (full manual)                                     |

## 18. One-page cheat sheet

| Item                  | Value                                                                                                         | Note                                |
|-----------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Board                 | 288 × 216 mm, 10 layers, 1.6 mm, ENIG, 50 Ω / 100 Ω                                                           | Progress Review v4, slide 16        |
| Devices               | DUT-A Intel 18A FCBGA 19 × 19 (compression socket); DUT-B QFN-40 (clamshell)                                  | slides 13, 14, 20                   |
| Input                 | 12 V DC, ≤ 0.88 A at the full envelope; eFuse TPS25990, limit 7 A                                             | derived                             |
| Core rail             | 0.75 V; 1.33 A expected / 4.0 A envelope; ±22.5 mV; remote sense at the balls                                 | assumed power; PD manual §10        |
| Core regulator limits | over-current ≈ 8 A; soft-start ≥ 5 ms; PMBus margining ±3 % / ±5 %                                            | derived                             |
| Other rails           | VDDIO 1.8 V (TPS62913); VDD_PLL 1.8 V (LT3045, need to ask); VAUX 3.3 V (TPS62912)                            | slide 17                            |
| PDN target            | 5.6 mΩ (envelope) / 16.9 mΩ (expected), 100 kHz – 50 MHz                                                      | derived                             |
| Decoupling per device | 2 × 470 μF, 8 × 47 μF, 16 × 4.7 μF, 24 × 0.1 μF (starting point)                                              | finalised by simulation             |
| Shunts                | VDD core 2 mΩ; VDDIO 100 mΩ; VDD_PLL 300 mΩ; VAUX 30 mΩ; 12 V 30 mΩ                                           | INA228, core on the ±40.96 mV range |
| Power-up order        | VAUX → VDDIO → VDD → VDD_PLL → clock → reset (≥ 100 ms later)                                                 | assumed; NA-04                      |
| Supervisor thresholds | VDD 0.69 V; VDDIO / VDD_PLL 1.67 V; VAUX 3.07 V                                                               | TPS3808 datasheet                   |
| Clock                 | XO + 4-output buffer (LVCMOS LMK1C1104 or an LVDS part — need to ask); SMA in / out; bypass ≤ 50 MHz assumed  | slide 18; NA-03                     |
| UART                  | 115,200 Bd 8N1; 8.68 μs/bit; 8,681 clocks/bit; 8.2 s per block; 10.9 min per batch up, 75 s down              | Architecture Manual §7.3, §17.3     |
| UART header           | FTDI TTL-232R-3V3 order; pin 3 unconnected (cable carries 5 V); pin 4 → chip RX; pin 5 ← chip TX              | FTDI datasheet                      |
| JTAG                  | 10-pin 1.27 mm Cortex-Debug order, VTref = 1.8 V, buffered                                                    | slide 18; assumed order             |
| Socket (assumed)      | 30 mΩ / 0.5 nH per pin; 30 + 30 power balls → 8 mV, 33 pH at 4 A                                              | vendor data pending                 |
| Thermal (assumed)     | θja 25 K/W still air / 12 K/W heatsink+airflow; Tj 71 / 133 °C still, 55 / 85 °C with heatsink at 1.2 / 3.7 W | package data pending                |
| Deliverables          | 4 boards, one iteration, 10-week plan; test sheet per board                                                   | slides 13, 22                       |
| Biggest open items    | ball map + sense balls (NA-01); PLL datasheet (NA-03); DUT-B identity (NA-11)                                 | chapter 16                          |